

Ridge Regression — Other Important Points

- **Used for:** Regression problems where overfitting or highly correlated features are present.
- **L2 Regularization:** Ridge uses the sum of squared coefficients as its penalty.
- **Multicollinearity:** Ridge works well when input features are highly correlated.
- **Hyperparameter:** alpha (or λ) controls the strength of regularization.
- **Standardization:** Features should usually be scaled before applying Ridge because the penalty depends on coefficient size.
- **Prediction:** After training, use `.predict()` just like Linear Regression.
- **Evaluation:** Common metrics are MAE, MSE, RMSE, and R^2 .
- **Scikit-learn:**

```
from sklearn.linear_model import Ridge
```

```
model = Ridge(alpha=1)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

Ridge Regression — Short Notes

1. **Coefficient Shrinkage:**
As λ increases, coefficients become smaller, but generally **not exactly 0**.
2. **Large coefficients:**
Large weights are penalized more because Ridge uses .
3. **Bias–Variance:**
 $\lambda \uparrow \rightarrow \text{Bias } \uparrow, \text{Variance } \downarrow$.
Too much $\lambda \rightarrow$ underfitting; too little $\rightarrow$ overfitting.
4. **Loss Function:**
5. **Key idea:**
Ridge = L2 regularization $\rightarrow$ penalizes large coefficients $\rightarrow$ reduces overfitting.
=====

Lasso Regression — Short Notes

1. What is Lasso?

Lasso Regression = Linear Regression + L1 Regularization

It adds a penalty based on the absolute value of coefficients:

2. Main Difference from Ridge

Ridge	Lasso
L2	L1
penalty	(
Shrinks coefficients	Shrinks coefficients
Usually doesn't make them 0	Can make them exactly 0
Doesn't perform feature selection	Performs feature selection

3. Feature Selection

Suppose:

Before Lasso:

Age → 5.2

Salary → 8.7

Experience → 0.0

Gender → 2.1

Because Experience has coefficient 0, Lasso effectively removes

Why Lasso creates zeros but Ridge doesn't

This is a very common ML interview question.

Lasso — L1

Lasso uses:

The important part is:

Its penalty has a constant slope, so as regularization increases, it can push a coefficient all the way to:

Once a coefficient becomes 0, that feature is effectively removed.

Therefore:

Lasso → exact zero coefficients → feature selection → sparse model

Ridge — L2

Ridge uses:

Its penalty depends smoothly on the size of w.

So Ridge keeps shrinking the coefficient:

$10 \rightarrow 5 \rightarrow 2 \rightarrow 0.5 \rightarrow 0.1 \rightarrow 0.01 \rightarrow \dots$

but normally doesn't make it exactly zero.

Therefore:

Ridge $\rightarrow$ coefficients approach zero but generally don't become zero.

Easy way to remember 🤖

Lasso $\rightarrow$ L1 $\rightarrow |w| \rightarrow$ can become 0 $\rightarrow$ Feature Selection

Ridge $\rightarrow$ L2 $\rightarrow w^2 \rightarrow$ shrinks toward 0 $\rightarrow$ Keeps features

Interview answer

Lasso creates sparsity because its L1 penalty can force coefficients exactly to zero, effectively performing feature selection. Ridge uses an L2 penalty that smoothly shrinks coefficients toward zero but generally does not make them exactly zero.

ElasticNet Regression — Short Notes

What is ElasticNet?

ElasticNet = Ridge + Lasso

It combines:

- L1 penalty $\rightarrow$ feature selection
- L2 penalty $\rightarrow$ coefficient shrinkage

Loss Function

Where:

- $\alpha \rightarrow$ overall regularization strength
- $r \rightarrow$ L1 ratio
- $r = 1 \rightarrow$ Lasso
- $r = 0 \rightarrow$ Ridge

Why use ElasticNet?

ElasticNet is useful when:

- There are many features
- Some features may be irrelevant
- Features have multicollinearity
- You want both feature selection + coefficient shrinkage

Example

Ridge:

$w_1 = 5, w_2 = 2, w_3 = 0.5$

Lasso:

$w_1 = 5, w_2 = 0, w_3 = 0$

ElasticNet:

$w_1 = 4, w_2 = 1, w_3 = 0$

So ElasticNet can shrink coefficients and remove some features.

Scikit-Learn

```
from sklearn.linear_model import ElasticNet
```

```
model = ElasticNet(alpha=1, l1_ratio=0.5)
```

```
model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)
```

 Remember

Ridge → L2 → Shrink

Lasso → L1 → Select features

ElasticNet → L1 + L2 → Shrink + Select
